

Math 237B Final

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Problem 1. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes such that $g \circ f$ is proper, g is quasi-projective and f is surjective. Prove or disprove that g is proper.

Solution. We'll prove that g is proper. We'll prove relevant statements in the following order:

- (a) g is separated, and f is proper.
- (b) g is of finite type.
- (c) g is universally closed.

Finally, use these results we can conclude that g is proper.

Proof of (a):

Since $g : Y \rightarrow Z$ is quasi-projective, there exists some scheme Y' , together with open immersion $\iota : Y \hookrightarrow Y'$ and projective morphism $\pi : Y' \rightarrow Z$, such that the following commutative diagram holds:

$$\begin{array}{ccc} Y & \xhookrightarrow{\iota} & Y' \\ g \downarrow & \swarrow \pi & \\ Z & & \end{array}$$

Which, recall that any open immersion is separated, and projective morphisms are proper (in particular separated), so $g = \pi \circ \iota$ is also separated.

As a consequence, since $g \circ f$ is proper, while g is separated, we concluded that f is proper also.

Proof of (b):

Let's show g is of finite type: Using the diagram in part (a), since $\pi : Y' \rightarrow Z$ is projective, it is proper (hence of finite type), so for any open affine subset $U \subseteq Z$ (say $U \cong \text{Spec}(A)$), such that $\pi^{-1}(U) = \bigcup_{i=1}^n V_i$, where each V_i is affine open subset, and $V_i \cong \text{Spec}(B_i)$ for some B_i that's finitely generated A -algebra.

Then, with open immersion $\iota : Y \hookrightarrow Y'$, one has $\iota^{-1}(\pi^{-1}(U)) = Y \cap \pi^{-1}(U) = \bigcup_{i=1}^n Y \cap V_i$. Since Y is open in Y' , each $Y \cap V_i$ is an open subset of V_i under its subspace topology, hence each $Y \cap V_i = \bigcup_{j \in J_i} D(f_{ij})$ for some $f_{ij} \in B_i$ (running over some index set J_i). Because each $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$ (which $(B_i)_{f_{ij}}$ is a finitely generated B_i -algebra), then with B_i being a finitely generated A -algebra, $(B_i)_{f_{ij}}$ is a finitely generated A -algebra also.

As a result, one has $g^{-1}(U) = \iota^{-1}(\pi^{-1}(U)) = \bigcup_{i=1}^n Y \cap V_i = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, where each $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$ has $(B_i)_{f_{ij}}$ being a finitely generated A -algebra.

Finally, since $g \circ f$ is finite type, it's quasi-compact, hence with affine open subset $U \subseteq Z$, one has $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$ being quasi-compact. However, since f is surjective, one has $g^{-1}(U) =$

$f(f^{-1}(g^{-1}(U)))$. Because it's an image of a quasi-compact set under continuous map, $g^{-1}(U)$ is quasi-compact. Hence, the previous cover for $g^{-1}(U)$, say $g^{-1}(U) = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, can be chosen as finite. This shows that g is of finite type.

Proof of (c):

To prove g is universally closed, for any morphism of schemes $h : Z' \rightarrow Z$ consider the following commutative diagram of multiple fibre products:

$$\begin{array}{ccc}
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 \downarrow f' & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 \downarrow g' & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

Which:

- $g' : Z' \times_Z Y \rightarrow Z'$ denotes the base change of $g : Y \rightarrow Z$ by $h : Z' \rightarrow Z$.
- $f' : (Z' \times_Z Y) \times_Y X \rightarrow Z' \times_Z Y$ denotes the base change of $f : X \rightarrow Y$ by $h' : Z' \times_Z Y \rightarrow Y$ (the pullback of $h : Z' \rightarrow Z$), which the surjectivity of f ensures the surjectivity of f' . And, since part (a) shows f is proper (which is universally closed), then f' is closed.

Now, we ought to show $(Z' \times_Z Y) \times_Y X$ together with $g' \circ f'$ and h'' satisfies the universal property of the fibre product $Z' \times_Z X$. For any scheme W together with two morphisms $p_1 : W \rightarrow X$ and $p_2 : W \rightarrow Z'$, such that $(g \circ f) \circ p_1 = h \circ p_2$. Then, the universal property of $Z' \times_Z Y$ guarantees a unique morphism of schemes $\varphi : W \rightarrow Z' \times_Z Y$, such that $p_2 = g' \circ \varphi$, and $f \circ p_1 = h' \circ \varphi$, as follow:

$$\begin{array}{ccc}
 W & \xrightarrow{p_1} & X \\
 \downarrow \exists! \varphi & & \downarrow f' \\
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 \downarrow f' & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 \downarrow g' & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

Furthermore, with the universal property of $(Z' \times_Z Y) \times_Y X$, the equality $f \circ p_1 = h' \circ \varphi$ guarantees a unique morphism of schemes $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$, such that $\varphi = f' \circ \psi$ and $p_1 = h'' \circ \psi$, as follow:

$$\begin{array}{c}
\begin{array}{ccc}
W & \xrightarrow{p_1} & X \\
\downarrow \exists! \psi & & \downarrow h'' \\
(Z' \times_Z Y) \times_Y X & \xrightarrow{\quad} & X \\
\downarrow f' & & \downarrow f \\
Z' \times_Z Y & \xrightarrow{h'} & Y \\
\downarrow g' & & \downarrow g \\
Z' & \xrightarrow{h} & Z
\end{array}
\end{array}$$

p_2 (curved arrow from W to Z')
 $\exists! \varphi$ (dotted arrow from W to $Z' \times_Z Y$)

This proves both the existence and the uniqueness of the morphism $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$, such that $p_2 = g' \circ \varphi = (g' \circ f') \circ \psi$, and $p_1 = h'' \circ \psi$ (as any $\psi' : W \rightarrow (Z' \times_Z Y) \times_Y X$ with such property is enforced to be equal to ψ , based on individual fibre product's property). So, $(Z' \times_Z Y) \times_Y X$ indeed satisfies the universal property of $Z' \times_Z X$, showing that $g' \circ f'$ is the base change of the morphism $(g \circ f) : X \rightarrow Z$. Which, since $g \circ f$ is proper, $g' \circ f'$ is closed.

Finally, with f' and $g' \circ f'$ being closed, and f' being surjective, we claim that g' is closed:

For any closed subset $V \subseteq Z' \times_Z Y$, by the continuity of f' , the subset $(f')^{-1}(V) \subseteq (Z' \times_Z Y) \times_Y X$ is closed. Hence, $(g' \circ f')((f')^{-1}(V)) \subseteq Z'$ is closed (by closeness of $g' \circ f'$).

Also, since f' is surjective, one simply has $f'((f')^{-1}(V)) = V$. Hence, the above set equality shows that $(g' \circ f')((f')^{-1}(V)) = g'(V) \subseteq Z'$ is closed. This proves that g' is closed.

With g' being based change of g over arbitrary morphism of schemes $h : Z' \rightarrow Z$, this shows that g is universally closed.

With all information above, we can conclude that g is separated, finite type, and universally closed, hence proper.

Problem 2. Let $f: X \rightarrow S$ and $g: Y \rightarrow S$ be projective morphisms. Prove or disprove that the projection map $X \times_S Y \rightarrow S$ is projective.

- (a) Base change of projective morphisms are projective.
- (b) Composition of projective morphisms are projective.

Proof of (a):

First, we'll show that base change preserves projective spaces: More precisely, given any $n \in \mathbb{N}$, with morphisms $g : Y \rightarrow S$, the projective n -space over S , $\pi : \mathbb{P}_S^n \rightarrow S$ has fibre product $Y \times_S \mathbb{P}_S^n$ satisfies the universal property of $\mathbb{P}_{Y}^n$.

Indeed, with the definition of $\mathbb{P}_S^n := S \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, we form the following commutative diagram:

$$\begin{array}{ccccc} Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\ \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\ Y & \xrightarrow{q} & S & \xrightarrow{s} & \mathrm{Spec}(\mathbb{Z}) \end{array}$$

Where, both small squares are fibre product diagrams. Also, the morphism $s \circ g : Y \rightarrow \mathrm{Spec}(\mathbb{Z})$ is the unique morphism based on the terminal property of $\mathrm{Spec}(\mathbb{Z})$ in the category of schemes.

Now, given any scheme W together with two morphisms $p_1 : W \rightarrow \mathbb{P}_{\mathbb{Z}}^n$ and $p_2 : W \rightarrow Y$, such that $\varphi \circ p_1 = (s \circ g) \circ p_2$, then the universal property of $\mathbb{P}_S^n$ generates a unique morphism $\psi : W \rightarrow \mathbb{P}_S^n$, such that $p_1 = s' \circ \psi$, and $g \circ p_2 = \pi \circ \psi$, as follow:

$$\begin{array}{ccccccc}
W & \xrightarrow{\quad p_1 \quad} & & & & & \\
& \searrow^{\exists ! \psi} & & & & & \\
Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n & & \\
\downarrow \pi' & & \downarrow \pi & & \downarrow \varphi & & \\
Y & \xrightarrow{q} & S & \xrightarrow{s} & \mathrm{Spec}(\mathbb{Z}) & &
\end{array}$$

Then, with the universal property of $Y \times_S \mathbb{P}_S^n$, the second equality $g \circ p_2 = \pi \circ \psi$ guarantees a unique morphism $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$, such that $\psi = g' \circ \rho$, and $p_2 = \pi' \circ \rho$, as follow:

$$\begin{array}{ccccccc}
W & & & & & & \\
\downarrow \exists! \rho & \searrow \exists! \psi & & & & & \\
Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n & & \\
\downarrow \pi' & & \downarrow \pi & & \downarrow \varphi & & \\
Y & \xrightarrow{g} & S & \xrightarrow{s} & \mathrm{Spec}(\mathbb{Z}) & & \\
\uparrow p_2 & & & & & & \\
W & \xrightarrow{p_1} & & & & &
\end{array}$$

Again, this proves both the existence and the uniqueness of the morphism $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$, such that $p_1 = s' \circ \psi = (s' \circ g') \circ \rho$, and $p_2 = \pi' \circ \rho$. Hence, $Y \times_S \mathbb{P}_S^n$ satisfies the universal property of $\mathbb{P}_Y^n := Y \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, showing $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$.

Now, we'll prove that base change preserves projective morphisms, again by the universality argument: Given a projective morphism $f : X \rightarrow S$, and another morphism $g : Y \rightarrow S$, we know there exists a suitable $n \in \mathbb{N}$, such that there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}_S^n$ and the projection $\pi : \mathbb{P}_S^n \rightarrow S$, such that $f = \pi \circ \iota$.

Which, based on the previous argument, we'll simply denote $Y \times_S \mathbb{P}_S^n$ as $\mathbb{P}_Y^n$. Together with the previous diagrams, we have the following:

$$\begin{array}{ccccc}
 X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n & \xrightarrow{g''} & X & & \\
 \downarrow \iota' & & \downarrow \iota & & \\
 \mathbb{P}_Y^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \downarrow \pi' & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Which, we took the fibre product $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$, and the morphism $\iota' : X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \hookrightarrow \mathbb{P}_Y^n$ is the pullback of the closed immersion $\iota : X \hookrightarrow \mathbb{P}_S^n$, hence ι' is also a closed immersion.

Now, we claim that it satisfies the universal property of $X \times_S Y$: In general, the above proof of preserving projective spaces shows that successive fibre product preserves fibre product in the full generality. By modifying the labels / objects a bit, we indeed get a proof that $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ satisfies the universal property of $X \times_S Y$ as fibre product, hence $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \cong X \times_S Y$.

Finally, by denoting $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ as $X \times_S Y$, the above diagram reduces to the following fibre product diagram:

$$\begin{array}{ccc}
 X \times_S Y & \xrightarrow{g''} & X \\
 \downarrow \iota' & & \downarrow \iota \\
 \mathbb{P}_Y^n & & \mathbb{P}_S^n \\
 \downarrow \pi' & & \downarrow \pi \\
 Y & \xrightarrow{g} & S
 \end{array}
 \quad \begin{array}{c} \curvearrowright f \end{array}$$

Which, the pullback of f is $\pi' \circ \iota' : X \times_S Y \rightarrow Y$, which is precisely a closed immersion into $\mathbb{P}_Y^n$, then a projection onto Y . This shows that the pullback of f is still projective, hence projective morphisms are preserved under base change.

Proof of (b):

Let $\rho : T \rightarrow U$ and $\varphi : U \rightarrow V$ be two projective morphisms, then there exists $n, m \in \mathbb{N}$, such that there exists open immersion $\iota_T : T \hookrightarrow \mathbb{P}_U^n$ and $\iota_U : U \hookrightarrow \mathbb{P}_V^m$, together with projections $\pi_{U,n} : \mathbb{P}_U^n \rightarrow U$ and $\pi_{V,m} : \mathbb{P}_V^m \rightarrow V$, such that the following diagram commutes:

$$\begin{array}{ccccc}
 \mathbb{P}_U^n & & \mathbb{P}_V^m & & \\
 \downarrow \iota_T & \searrow \pi_{U,n} & \downarrow \iota_U & \searrow \pi_{V,m} & \\
 T & \xrightarrow{\rho} & U & \xrightarrow{\varphi} & V
 \end{array}$$

Now, let's use some information from part (a): Recall that given any morphism $Y \rightarrow S$ and the projection $\mathbb{P}_S^n \rightarrow S$, the fibre product $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$ (and it's true for arbitrary Y, S and $n \in \mathbb{N}$, as the proof is universal); then, from the above diagram, consider $\varphi : U \rightarrow V$ as the composition $U \xrightarrow{\iota_U} \mathbb{P}_V^m \xrightarrow{\pi_{V,m}} V$, and consider the projection $\pi_{V,n} : \mathbb{P}_V^n \rightarrow V$. Notice by the previous statement, we can construct the projective space $\mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$, as follow:

$$\begin{array}{ccccc} & & \mathbb{P}_{\mathbb{P}_V^m}^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n \\ & & \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} \\ U & \xhookrightarrow{\iota_U} & \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V \end{array}$$

Which, the same argument shows that $\mathbb{P}_U^n \cong U \times_{\mathbb{P}_V^m} \mathbb{P}_{\mathbb{P}_V^m}^n$, as follow:

$$\begin{array}{ccccc} \mathbb{P}_U^n & \xhookrightarrow{(\iota_U)'} & \mathbb{P}_{\mathbb{P}_V^m}^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n \\ \downarrow \pi_{U,n} & & \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} \\ U & \xhookrightarrow{\iota_U} & \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V \end{array}$$

Which, notice that $(\iota_U)' : \mathbb{P}_U^n \hookrightarrow \mathbb{P}_{\mathbb{P}_V^m}^n$ is a base change of the closed immersion $\iota_U : U \hookrightarrow \mathbb{P}_V^m$, hence is still a closed immersion; composing with the closed immersion $\iota_T : T \hookrightarrow \mathbb{P}_U^n$, we get a closed immersion $(\iota_U)' \circ \iota_T : T \hookrightarrow \mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$, and with the projections $\pi_{V,n} \circ (\pi_{V,m})' : \mathbb{P}_{\mathbb{P}_V^m}^n \rightarrow V$, they compose to be the original morphism $T \rightarrow U \rightarrow V$.

Hence, to prove the composition $\varphi \circ \rho : T \rightarrow V$ is projective, it suffices to show that $\mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$ embeds into some $\mathbb{P}_V^N$, and the projection backs to $\mathbb{V}$ recovers $\varphi \circ \rho$.

Let's first show that there exists a morphism $p : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$. Consider the following diagram:

$$\begin{array}{ccccc} \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n & \longrightarrow & \mathbb{P}_{\mathbb{Z}}^n \\ \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} & & \downarrow \\ \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V & \searrow & \\ \downarrow & & & & \downarrow \\ \mathbb{P}_{\mathbb{Z}}^m & \longrightarrow & \text{Spec}(\mathbb{Z}) & & \end{array}$$

Which, with each „square“ (two are deformed a bit) being fibre squares, the above diagram commutes. Hence, by the universal property of $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ guarantees a unique morphism $P : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$.

Now, assume we know the existence of Segre Embedding for projective spaces of $\mathbb{Z}$ (suggested in Hartshorne), say $s : \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \hookrightarrow \mathbb{P}_{\mathbb{Z}}^{nm+n+m}$, the with $N := nm + n + m$ consider the following commutative diagram:

$$\begin{array}{ccc} \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{p} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\ \downarrow \exists! s' & & \downarrow s \\ \mathbb{P}_V^N & \longrightarrow & \mathbb{P}_{\mathbb{Z}}^N \\ \downarrow \pi_{V,N} & & \downarrow \\ V & \longrightarrow & \text{Spec}(\mathbb{Z}) \end{array}$$

$\pi_{V,n} \circ (\pi_{V,m})' \searrow$

Then, the universality of $\mathbb{P}_V^N = V \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^N$ guarantees a unique morphism $s' : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_V^N$. Finally, we claim that s' is a closed immersion:

Notice in the above diagram, the tall rectangle involving $\mathbb{P}_V^m \times_V \mathbb{P}_V^n, \mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n, V, \text{Spec}(\mathbb{Z})$ forms a fibre square by diagram chase.

Then, this in fact enforces $\mathbb{P}_V^m \times_V \mathbb{P}_V^n \cong \mathbb{P}_V^N \times_{\mathbb{P}_Z^N} (\mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n)$: Indeed, given any scheme W together with two morphisms $w_1 : W \rightarrow \mathbb{P}_V^N$ and $w_2 : W \rightarrow \mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n$, such that the following holds:

$$\begin{array}{ccccc}
 W & & & & \\
 \swarrow \exists! w & & & & \searrow w_2 \\
 & \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{p} & \mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n & \\
 \downarrow \exists! s' & & & \downarrow s & \\
 \mathbb{P}_V^N & \xrightarrow{\quad} & & \mathbb{P}_Z^N & \\
 \downarrow \pi_{V,N} & & & \downarrow & \\
 V & \xrightarrow{\quad} & & \text{Spec}(\mathbb{Z}) &
 \end{array}$$

(Note: In the original image, there is a curved arrow from W to $\mathbb{P}_V^N$ labeled w_1 .)

Which, the existence and uniqueness of $w : W \rightarrow \mathbb{P}_V^m \times_V \mathbb{P}_V^n$ is guaranteed by the tall fibre rectangle, showing the top square is a fibre square. Since this characterizes $\mathbb{P}_V^m \times_V \mathbb{P}_V^n$ as the fibre product $\mathbb{P}_V^N \times_{\mathbb{P}_Z^N} (\mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n)$, then $s' : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_V^N$ is a base change of $s : \mathbb{P}_Z^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_Z^n \hookrightarrow \mathbb{P}_Z^N$ (by the morphism $\mathbb{P}_V^N \rightarrow \mathbb{P}_Z^N$), showing s' is a closed immersion (since s is).

Finally, we get this massive commutative diagram:

$$\begin{array}{ccccccc}
 T & \xrightarrow{\iota_T} & \mathbb{P}_U^n & \xrightarrow{(\iota_U)'} & \mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{s'} & \mathbb{P}_V^N \\
 \downarrow \rho & \swarrow \pi_{U,n} & & \swarrow (\pi_{V,n})' & & & \\
 U & \xrightarrow{\iota_U} & \mathbb{P}_V^m & & & & \\
 \downarrow \varphi & \swarrow \pi_{V,m} & & \swarrow \pi_{V,N} & & & \\
 V & & & & & &
 \end{array}$$

(Note: In the original image, there is a curved arrow from T to V labeled $\varphi \circ \rho$.)

Which, $\iota_T, (\iota_U)', s'$ are all constructed as closed immersions, hence the composition $T \hookrightarrow \mathbb{P}_V^N$ is also a closed immersion. This realizes $\varphi \circ \rho : T \rightarrow V$ as a projective morphism, hence composition of projective morphisms are still projective.

Finally, back to the original problem, given $f : X \rightarrow S$ and $g : Y \rightarrow S$ two projective morphisms, (a) guarantees their pullbacks to be projective morphisms also. So, $f' : X \times_S Y \rightarrow Y$ and $g' : X \times_S Y \rightarrow X$ are projective morphisms, satisfying the following fibre square:

$$\begin{array}{ccc}
 X \times_S Y & \xrightarrow{f'} & Y \\
 g' \downarrow & & \downarrow g \\
 X & \xrightarrow{f} & S
 \end{array}$$

Then, (b) guarantees the composition of projective morphisms to be projective, hence $f \circ g' = g \circ f' : X \times_S Y \rightarrow S$ is projective, finishing our statement.

3 D

Problem 3. Let $f : X \rightarrow Y$ be a morphism of separated scheme of finite type over a Noetherian scheme S . Let $Z \subset X$ be a closed subscheme which is proper over S . Show that $f(Z)$ is closed in Y .

Solution. To show $f(Z) \subseteq Y$ is closed, we'll show that with the closed immersion $\iota : Z \hookrightarrow X$, one has $f \circ \iota : Z \rightarrow Y$ is proper, which will show $f \circ \iota(Z) = f(Z)$ is closed in Y .

First, let $\varphi_X : X \rightarrow S$ and $\varphi_Y : Y \rightarrow S$ be the morphism that define X, Y as separated schemes of finite type over S , this implies both φ_X, φ_Y are of finite type and separated morphisms.

Also, consider the closed immersion $\iota : Z \hookrightarrow X$, since Z is proper over S , we have this morphism $\varphi_X \circ \iota : Z \rightarrow S$ be proper.

Now, since $f : X \rightarrow Y$ is a morphism of schemes over S , we have the following commutative diagram:

$$\begin{array}{ccccc}
 & & Z & & \\
 & & \searrow & & \\
 & & \iota & & \\
 & & \searrow & & \\
 & X & \xrightarrow{f} & Y & \\
 & \searrow & & \swarrow & \\
 & \varphi_X & & \varphi_Y & \\
 & & S & &
 \end{array}$$

Or, $\varphi_X = \varphi_Y \circ f$, which implies $\varphi_X \circ \iota = \varphi_Y \circ (f \circ \iota)$. Since $\varphi_X \circ \iota$ is proper, with φ_Y being separated, we have $f \circ \iota$ being proper also (which is a closed map). With Z being closed in itself, $f \circ \iota(Z) = f(Z) \subseteq Y$ is closed. This finishes our proof.

4 ND ((a) remaining)

Problem 4. Let k be a field and let $R = k[X, Y, Z]/I$, where $I = (X - YZ, XZ - Y^2)$. Let $W = \text{Spec}(R)$.

- (a) Is W a reduced scheme? Justify your answer.
- (b) Is W irreducible? If not, what are its irreducible components?
- (c) Prove or disprove that W has infinitely many rational points if k is infinite (recall that a point $x \in W$ is called rational point if the canonical map $k \rightarrow k(x)$ is an isomorphism, where $k(x)$ is the residue field of the local ring $\mathcal{O}_{W,x}$).

Solution. For the sake of working through all subproblems, we'll first do some calculations regarding two ideals of $k[X, Y, Z]$, defined by $P = (X, Y)$ and $Q = (X - Z^3, Y - Z^2)$. In particular, we claim that they're prime ideals that contains I :

- For $P = (X, Y)$, it's clear that $k[X, Y, Z]/P \cong k[Z]$, hence it's prime; one the other hand, one has $X, Y \in P$ implies $YZ, XZ, Y^2 \in P$, hence $X - YZ, XZ - Y^2 \in P$, showing $I \subseteq P$.
- For Q , we'll show that $k[X, Y, Z]/Q \cong k[T]$ also: Consider the ring homomorphism $\varphi : k[X, Y, Z] \rightarrow k[T]$ by $\varphi(f(X, Y, Z)) = f(T^3, T^2, T)$. Which, this morphism is surjective, as any polynomial $f(T) \in k[T]$ also appears in $k[X, Y, Z]$ (say denotes as $f(Z)$), then $\varphi(f(Z)) = f(T)$. This shows that $k[X, Y, Z]/\ker(\varphi) \cong k[T]$.

Now, we show that $Q = \ker(\varphi)$: For the generators of Q , we have $\varphi(X - Z^3) = T^3 - (T)^3 = 0$, and $\varphi(Y - Z^2) = T^2 - (T)^2 = 0$, hence $Q \subseteq \ker(\varphi)$ (as its generators are contained in ther kernel).

For the converse, we claim that all polynomials can be expressed in the form $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ for some $f, g \in k[X, Y, Z]$, and $h(Z) \in k[Z]$: Given any monomial $X^n Y^m Z^p$, we can rewrite it as the following:

$$\begin{aligned} X^n Y^m Z^p &= ((X - Z^3) + Z^3)^n ((Y - Z^2) + Z^2)^m Z^p \\ &= \left(\sum_{k=0}^n \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k} \right) \left(\sum_{l=0}^m \binom{m}{l} (Y - Z^2)^l (Z^2)^{m-l} \right) Z^p \\ &= \left(Z^{3n} + \sum_{k=1}^n \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k} \right) \left(Z^{2m} + \sum_{l=1}^m (Y - Z^2)^l (Z^2)^{m-l} \right) Z^p \end{aligned} \quad (4.1)$$

Which, define $f := \sum_{k=1}^n \binom{n}{k} (X - Z^3)^{k-1} (Z^3)^{n-k}$, and $g := \sum_{l=1}^m (Y - Z^2)^{l-1} (Z^2)^{m-l}$, the above can be rewritten as follow:

$$\begin{aligned} X^n Y^m Z^p &= (Z^{3n} + f \cdot (X - Z^3))(Z^{2m} + g \cdot (Y - Z^2))Z^p \\ &= (fg(Y - Z^2)Z^p) \cdot (X - Z^3) + (gZ^{3n+p}) \cdot (Y - Z^2) + Z^{3n+2m+p} \end{aligned} \quad (4.2)$$

This shows that each monomial can be written as some $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + Z^q$. Hence, with all polynomials being finite k -linear combinations of the monomials, they can all be written in the form of $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ for $f, g \in k[X, Y, Z]$ and $h(Z) \in k[Z]$.

As a consequence, for any polynomial $\ell(X, Y, Z) \in \ker(\varphi)$, since there exists $f, g \in k[X, Y, Z]$ and $h(Z) \in k[Z]$, such that $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$, then the evaluation by φ gives the following:

$$\varphi(\ell(X, Y, Z)) = \varphi(f \cdot (X - Z^3)) + \varphi(g \cdot (Y - Z^2)) + \varphi(h(Z)) = h(T) = 0 \quad (4.3)$$

(Note: recall that $X - Z^3, Y - Z^2 \in \ker(\varphi)$ are verified). Hence, one must have $h(T) = 0$, or $h(Z) = 0$, showing that $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) \in Q$. This proves that $\ker(\varphi) \subseteq Q$, finishing the proof of $\ker(\varphi) = Q$.

Then, since $k[X, Y, Z]/Q \cong k[T]$ based on the homomorphism φ , then Q is a prime ideal.

Finally, to say $I \subseteq Q$, use the algorithm proposed above, some calculation shows the following:

$$\begin{aligned}
 \bullet \quad X - YZ &= X - ((Y - Z^2) + Z^2)Z = (X - Z^3) + Z(Y - Z^2) \in Q & (4.4) \\
 \bullet \quad XZ - Y^2 &= XZ - ((Y - Z^2) + Z^2)^2 \\
 &= XZ - ((Y - Z^2)^2 + 2Z^2(Y - Z^2) + Z^4) \\
 &= XZ - Z^4 - ((Y - Z^2) + 2Z^2)(Y - Z^2) \\
 &= Z(X - Z^3) + (Y + Z^2)(Y - Z^2) \in Q
 \end{aligned} \tag{4.5}$$

Hence, the generators of I are in Q , showing $I \subseteq Q$.

Now, since $W = \text{Spec}(R) \cong V(I)$ as closed subschemes of $\text{Spec}(k[X, Y, Z])$, we'll work in this regime.
Here are the solutions to the subproblems:

(a) **Not done yet.**

(b) We show that W is reducible, in particular that $W = V(I) = V(P) \cup V(Q)$, where $V(P), V(Q)$ are its irreducible components.

First, since $I \subseteq P$ and $I \subseteq Q$, it's clear that $V(P), V(Q) \subseteq V(I)$, hence $V(P) \cup V(Q) \subseteq V(I)$; to claim the reverse inclusion, we'll consider the following element:

$$Z(X - YZ) - (XZ - Y^2) = XZ - YZ^2 - XZ + Y^2 = Y(Y - Z^2) \in I \tag{4.6}$$

Suppose $K \in V(I)$ is a prime ideal containing I , then $Y(Y - Z^2) \in K$, which the prime property guarantees $Y \in K$ or $(Y - Z^2) \in K$.

- If $Y \in K$, then since $X - YZ \in I \subseteq K$, one has $X = (X - YZ) + YZ \in K$, showing $P = (X, Y) \subseteq K$, or $K \in V(P)$.
- Else if $Y - Z^2 \in K$, then since $X - YZ \in I \subseteq K$ again, one has $X - Z^3 = (X - YZ) + (YZ - Z^3) = (X - YZ) + Z(Y - Z^2) \in K$, showing $Q = (X - Z^3, Y - Z^2) \subseteq K$, or $K \in V(Q)$.

Hence, in either case $K \in V(P) \cup V(Q)$, showing $W = V(I) = V(P) \cup V(Q)$.

To show that W is reducible, we'll show that $V(P), V(Q)$ are proper closed subsets of W : Suppose the contrary that one of them is not proper closed subset of W , there are two cases:

- If $V(P) = W$, this shows that $V(Q) \subseteq V(P)$; with P, Q being prime (which both are radicals), this shows that $Q \supseteq P$. However, if consider the ring homomorphism $\psi : k[X, Y, Z] \rightarrow k$ by $\psi(X, Y, Z) = \psi(1, 1, 1)$, notice that $\psi(X - Z^3) = 1 - 1^3 = 0$ and $\psi(Y - Z^2) = 1 - 1^2 = 0$, showing $Q = (X - Z^3, Y - Z^2) \subseteq \ker(\psi)$; yet, if plugin the generators of P , we get $\psi(X) = \psi(Y) = 1 \neq 0$, showing $P \not\subseteq \ker(\psi)$, which contradicts our construction that $\ker(\varphi) \supseteq Q \supseteq P$.
- Else if $V(Q) = W$, this shows that $V(P) \subseteq V(Q)$, which implies $P \supseteq Q$. However, consider the ring homomorphism $\rho : k[X, Y, Z] \rightarrow k$ by $\rho(f(X, Y, Z)) = f(0, 0, 1)$, notice that $\rho(X) = \rho(Y) = 0$, showing $P = (X, Y) \subseteq \ker(\rho)$; yet, if plugin the generators of Q , we get $\rho(X - Z^3) = 0 - 1^3 = -1 \neq 0$, showing $Q \not\subseteq \ker(\rho)$, again this contradicts our construction that $\ker(\rho) \supseteq P \supseteq Q$.

Since in either case we get a contradiction, then $V(P), V(Q)$ cannot be the whole W , showing they're proper closed subsets, which proves that $W = V(P) \cup V(Q)$ is reducible.

Finally, to show $V(P), V(Q)$ are its irreducible components, consider any other irreducible subsets $W' \subseteq W = V(I)$, since W' is also closed in $\text{Spec}(k[X, Y, Z])$ (by the closeness of $V(I)$ in it), then $W' = V(K)$ for some ideal $K \subseteq k[X, Y, Z]$. Which, the irreducibility of W' guarantees K to be a prime ideal. However, if $V(K) \subseteq V(I)$, one has $I \subseteq \sqrt{I} \subseteq K$ (by primeness, it is a radical), showing that $K \in V(I) = V(P) \cup V(Q)$. This implies that either $K \in V(P)$ ($P \subseteq K$, or $V(P) \supseteq V(K)$), or $K \in V(Q)$ ($Q \subseteq K$, or $V(Q) \supseteq V(K)$), showing that $W' = V(K)$ is contained in $V(P)$ or $V(Q)$. Hence, these two must be the irreducible components of W .

- (c) With k being infinite, we'll show that there are infinitely rational points in W . For any $t \in k$, consider the maximal ideal $\mathfrak{m}_t := (X, Y, Z - t) \subset k[X, Y, Z]$. Notice that it contains I , since $X - YZ, XZ - Y^2 \in \mathfrak{m}_t$ by the fact that $X, Y \in \mathfrak{m}_t$. Hence, $x := \mathfrak{m}_t$ can be realized as a maximal ideal in $R = k[X, Y, Z]/I$ by descending to the quotient, which is realized as a closed point in $W = \text{Spec}(R)$.

To compute its residue field, let $\overline{\mathfrak{m}}_t \subseteq R$ denotes the quotient of $\mathfrak{m}_t$. Then, its corresponding residue field is as follow:

$$k(\overline{\mathfrak{m}}_t) = \frac{R_{\overline{\mathfrak{m}}_t}}{\overline{\mathfrak{m}}_t R_{\overline{\mathfrak{m}}_t}} \cong \left(\frac{R}{\overline{\mathfrak{m}}_t} \right)_{\overline{\mathfrak{m}}_t} \cong k \quad (4.7)$$

Note that since $I \subseteq \mathfrak{m}_t$, then the composition of projections $\varphi : k[X, Y, Z] \twoheadrightarrow R \twoheadrightarrow R/\overline{\mathfrak{m}}_t$ has the kernel being precisely $\varphi^{-1}(\overline{\mathfrak{m}}_t) = \mathfrak{m}_t$, hence $k \cong k[X, Y, Z]/\mathfrak{m}_t \cong R/\overline{\mathfrak{m}}_t$ (where, the first isomorphism can be realized by the evaluation $e : k[X, Y, Z] \twoheadrightarrow k$ by $e(f(X, Y, Z)) = f(0, 0, t)$, where the kernel is precisely $\mathfrak{m}_t = (X, Y, Z - t)$).

Hence, with k being infinite, there are infinitely many $t \in k$, with $\overline{\mathfrak{m}}_t \subseteq R$ being a maximal ideal (or a closed point in $W = \text{Spec}(R)$), such that the residue field is isomorphic to k . Hence, W has infinitely many rational points.

5 ND (finish the open subscheme part)

Problem 5. Let $f : X \rightarrow Y$ be a surjective morphism between two schemes which are finite type and separated integral schemes over a field. Assume that X is an affine scheme. Prove or disprove that Y is an affine scheme if and only if it is a quasi-affine scheme (recall that quasi-affine means open subscheme of an affine scheme).

Solution. We'll prove that Y is affine $\iff Y$ is quasi-affine.

$\implies$:

Suppose Y is affine, then it's an open subscheme of Y (an affine scheme), hence Y is quasi-affine.

$\impliedby$:

Suppose that Y is quasi-affine. Here, we'll denote the global section of Y as $A := \mathcal{O}_Y(Y)$, and let B be a commutative ring satisfies $X \cong \text{Spec}(B)$ (which, X, Y being integral schemes implies A, B are integral domains). Let k be a field where X, Y are finite type and separated integral schemes over it. Which, given $\varphi_X : X \rightarrow \text{Spec}(k)$, $\varphi_Y : Y \rightarrow \text{Spec}(k)$ as the corresponding morphisms, φ_X, φ_Y are finite type and separated morphisms that satisfy the following diagram:

$$\begin{array}{ccc} X = \text{Spec}(B) & \xrightarrow{f} & Y \\ & \searrow \varphi_X & \swarrow \varphi_Y \\ & \text{Spec}(k) & \end{array}$$

Then, on the global section level, this reverses to the following diagram of ring homomorphisms:

$$\begin{array}{ccc} B & \xleftarrow{f^\#} & \mathcal{O}_Y(Y) = A \\ & \swarrow \varphi_X^\# & \searrow \varphi_Y^\# \\ & k & \end{array}$$

Now, we claim the following statements in order:

- (a) Y can be realized as open subschemes of $\text{Spec}(A)$.
- (b) Y is affine, in particular $Y \cong \text{Spec}(A)$.

Then, we'll use these to patch up more results.

Proof of (a):

Since Y is assumed to be quasi-affine, there exists an open immersion $\iota_Y : Y \hookrightarrow \text{Spec}(R)$ for some commutative ring R . Then, since the affinization of Y , say $\text{Aff}_Y = \text{Spec}(A)$, with the natural morphism of schemes $i_Y : Y \rightarrow \text{Aff}_Y$, there exists a unique morphism of schemes $\psi_Y : \text{Aff}_Y \rightarrow \text{Spec}(R)$, such that the following commutative diagram holds:

$$\begin{array}{ccc} Y & \xrightarrow{i_Y} & \text{Aff}_Y \\ & \searrow \iota_Y & \swarrow \exists! \psi_Y \\ & \text{Spec}(R) & \end{array}$$

Which, since ι_Y is injective (as set map), this enforces i_Y to also be injective also.

Now, consider the ring homomorphism $\psi_Y^\# : R \rightarrow A$. For any $P \in Y \hookrightarrow \text{Spec}(R)$, there exists some element $f \in R$, such that $P \in D(f) \subseteq Y$ (since Y is open subscheme of $\text{Spec}(R)$). Then, denote $f' := \varphi_Y^\#(f) \in A$. Using the Hartshorne Notation, we can consider the subset $Y_{f'} \subseteq Y$ (denoting all points $y \in$

Y , such that the localization $f'_y \in \mathcal{O}_{Y,y}$ is invertible, or not in the maximal ideal $\mathfrak{m}_y \subseteq \mathcal{O}_{Y,y}$. Notice that this set is precisely $Y_{f'} = D(f)$: Given the ring homomorphism $\varphi_Y^\# : R \rightarrow A$ that sends $\varphi_Y^\#(f) = f'$, then compose with the localization, ...

as a result, one has $R_f \cong \mathcal{O}_{\text{Spec}(R)}(D(f)) = \mathcal{O}_Y(D(f)) = \mathcal{O}_Y(Y_{f'}) = \mathcal{O}_Y(D(f')) \cong A_{f'}$, showing that $P \in D(f) \subseteq Y$ has an affine neighborhood $D(f) \cong D(f') \cong \text{Spec}(A_{f'})$ in Y .

This shows that $i_Y : Y \hookrightarrow \text{Spec}(A)$ is actually an open immersion (as any point in Y has an open neighborhood in Y , such that it's scheme-isomorphic to a fundamental open subset of $\text{Spec}(A)$). So, Y is an open subscheme of $\text{Spec}(A)$.

Proof of (b):

Now, together with the open immersion $i_Y : Y \hookrightarrow \text{Spec}(A)$, the morphism $\varphi_Y : Y \rightarrow \text{Spec}(k)$ uniquely factors through some morphism $\rho : \text{Spec}(A) \rightarrow \text{Spec}(k)$ (since $\text{Spec}(A) = \text{Aff}_Y$ the affinization), which we have the following commutative diagram of morphisms:

$$\begin{array}{ccccc} X & \xrightarrow{f} & Y & \xrightarrow{i_Y} & \text{Spec}(A) \\ & \searrow \varphi_X & \downarrow \varphi_Y & \swarrow \exists! \rho & \\ & & \text{Spec}(k) & & \end{array}$$

Which, on the level of global section, we have the following commutative diagram:

$$\begin{array}{ccccc} B & \xleftarrow{f^\#} & A & \xleftarrow{\text{id}_A} & A \\ & \searrow \varphi_X^\# & \uparrow \varphi_Y^\# & \nearrow \rho^\# & \\ & & k & & \end{array}$$

Where the middle A is given by the global section of Y , $A = \mathcal{O}_Y(Y)$.

Then, this implies that the morphism $i_Y \circ f : X = \text{Spec}(B) \rightarrow \text{Spec}(A)$ is induced by the ring homomorphism $f^\# \circ \text{id}_A = f^\# : A \rightarrow B$; and, for the diagram to commute, $f^\#$ is in fact a k -algebra homomorphism.

Now, notice that $i_Y \circ f$ is a dominant morphism: Because A is an integral domain, then $\text{Spec}(A)$ is irreducible, hence $Y \subseteq \text{Spec}(A)$ as a open subscheme is dense and irreducible (or $\overline{Y} = \text{Spec}(A)$). With f being surjective, one has $f(X) = Y$, hence $\overline{i_Y \circ f(X)} = \overline{i_Y(Y)} = \overline{Y} = \text{Spec}(A)$.

Finally, we claim that $Y = \text{Spec}(A)$: Suppose not, then since $\text{Spec}(A) \setminus Y$ is closed nonempty subset (by openness of Y), there exists ideal $I \subseteq A$, such that $V(I) = \text{Spec}(A) \setminus Y$ (and I must be proper for $V(I)$ to be nonempty). Now, choose a maximal ideal $\mathfrak{m} \subseteq A$ such that $I \subseteq \mathfrak{m}$, then $\mathfrak{m} \in V(I)$ is a point not containing in the image of $i_Y \circ f$ by our choice.

However, if consider $f^\#(\mathfrak{m})B$ as an idea